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School of Electrical Engineering & Computer Science

Semester Two Examinations, 2023

ELEC7309 Power System Planning and Reliability

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

•This is a Closed Book examination - no written materials permitted
•Casio FX82 series or UQ approved and labelled calculator only
•During Planning Time - Students are encouraged to review and plan responses to the exam questions
•This examination paper will be released to the Library

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

None

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

2 x 14-Page Answer Booklet
1 x 1cm x 1cm Graph Paper

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

There are two Sections in this exam (PART-A and PART-B). Total Marks in the exam is 60%. Answer each section ((PART-A and PART-B)) in a separate Booklet.

For Examiner Use Only

Question	Mark

Total _____

PART A: POWER SYSTEM PLANNING
ANSWER ALL THREE QUESTIONS FROM PART A

Question A1 **(Total marks 10)**

(a) What is the difference between decentralised peer-to-peer and community-based peer-to-peer trading markets? **(3 Marks)**

(b) Examine a peer-to-peer marketplace where 10 consumers and 6 prosumers participate. The functioning of this marketplace entails giving priority to consumers with the highest demand and prosumers with the greatest surplus. However, if multiple consumers and prosumers bid for the same amount, they are served according to their bidding sequence (first bid first serve). A mid-market rule is employed to establish the price for peer-to-peer transactions. The energy auction bids for various consumers are detailed in Table 1, while those for prosumers are presented in Table 2. Compute the earnings and expenses of individual customers based on their available supply and demand during the chosen trading period.

It's important to note that during this period, electricity procurement from the grid costs 30 cents per kWh, and the feed-in tariff rate stands at 7 cents per kWh. It's assumed that none of the customers possess energy storage capabilities, and all consumers must satisfy their demand requirements. **(7 marks)**

Table 1 for Question A1 (b): Demand of various consumers.

Bidding sequence	1	2	3	4	5	8	6	8	9	10
Consumer	2	5	8	7	1	3	4	6	10	9
Demand (kWh)	5	3	4	4	2	6	6.5	5.4	2.1	6

Table 2 for Question A1 (b): Supply from various prosumers.

Bidding sequence	1	2	3	4	5	6
Prosumer	5	3	1	2	6	4
Surplus (kWh)	7	5	6	5.5	7.5	6

Question A2 **(Total marks 10)**

(a) You have been given a task by GHD to determine the feasibility of a hydro-plant by calculating the BCR (Benefit/Cost Ratio) using the information given in Table 3. According to GHD's policy, they do not recommend any project unless the estimated benefit is at least 1.5 times higher than the cost. Based on your analysis, what is your recommendation about the feasibility of the project (feasible or not feasible)? Why? **(3 marks)**

Table 3 for Question A2 (a)

Total investment	$ 7000k
Lifetime	45 years
Annual operation cost	$ 150k
Annual earnings	$ 800k
Discount rate	8%

(b) Consider the system shown in Figure 1. The marginal cost π_A of region A and π_B of region B are given by the following two equations:

$$\pi_A = 15 + 0.05\, P_A \text{ (in \$/MWh)}$$
$$\pi_B = 17 + 0.06\, P_B \text{ (in \$/MWh)}$$

Assume that the systems are connected without any limit on transfer capacity. Calculate the electricity price and the electricity cost per hour for the customers in each system. If the Merchant Transmission wants to maximise the revenue, what should be the flow capacity (F) of the interconnection line? **(7 marks)**

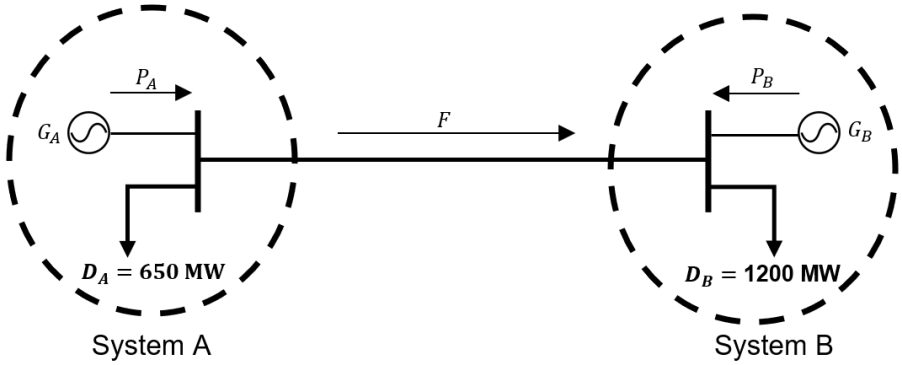


Figure 1 for Question A2 (b).

Question A3 **(Total marks 10)**

For three generation facilities, P, Q, and R, with details given in Table 4, assume an interest rate of 8% and the possibility of choosing any generation capacity of the above-mentioned technologies (P, Q, and R).

Table 4 for Question A3

Technology type	Investment cost ($/kW)	Plant life (years)	Fuel cost ($/MWh)
P	1,500	40	25
Q	500	28	35
R	150	20	55

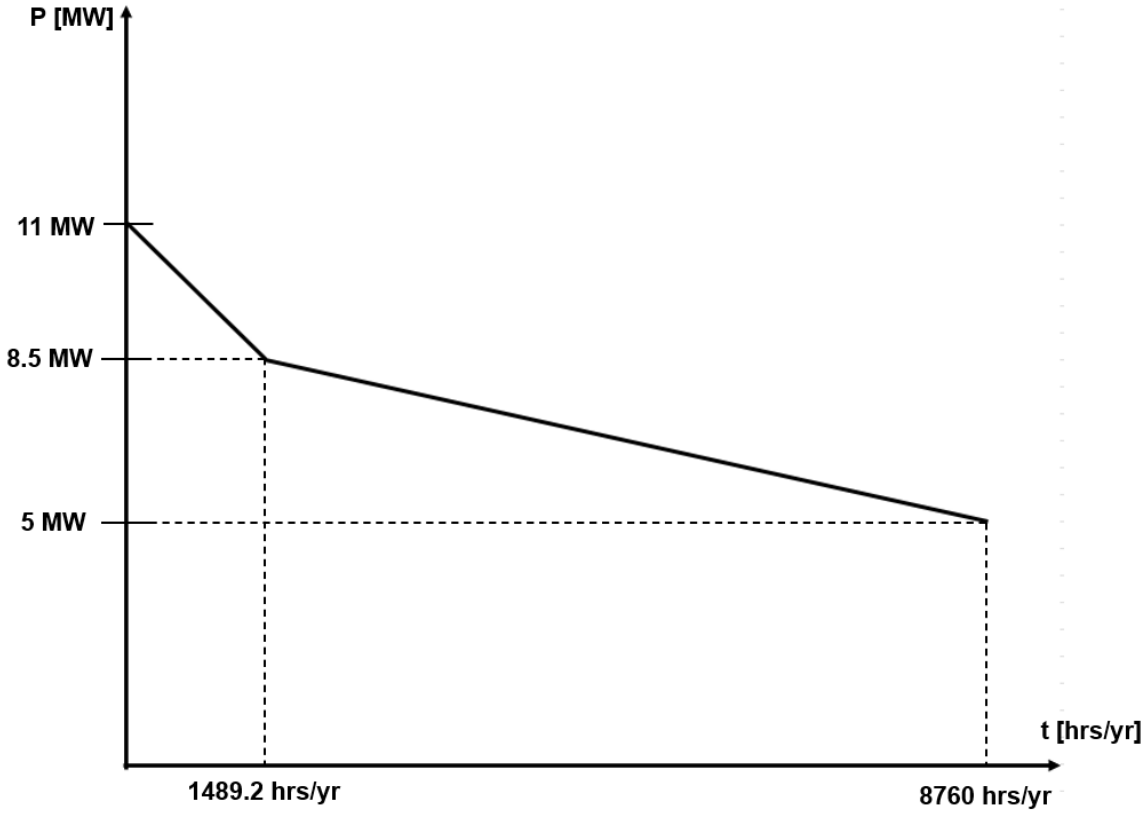


Figure 2 for Question 3

a) Derive the equations of annual cost as a function of capacity factor for each of the technologies using the data given in Table 4 and the Load Duration Curve shown in Figure 2. **(3 marks)**

b) Using the screening curve technique, determine the optimal type and capacity to supply the demand characteristics by the Load Duration Curve shown in Figure 2. **(3 marks)**

c) Based on the results obtained in b), calculate the annual operating cost for generation type Q. **(4 marks)**

PART B: POWER SYSTEM RELIABILITY QUESTIONS ARE IN NEXT PAGE

PART B: POWER SYSTEM RELIABILITY
ANSWER ALL THREE QUESTIONS FROM PART B

Question B1 **(Total marks 10)**

A system contains three identical 50 MW generating units (A, B and C) each with a 5% FOR and one 100 MW unit with a 6% FOR. The system peak load for a specified period is 200 MW.

(a) Calculate the LOLE and the expected load loss for this system. Assume that the appropriate load characteristic is a straight line from 100% to 60% points. **(5 marks)**

(b) An additional unit of 50 MW is available for addition to the system with a FOR of 4%. The load growth in the system is 10%. Which year the generating unit should be added to the network to keep the LOLE either equal or less than that found in (a)? **(5 marks)**

Question B2 **(Total marks 10)**

Consider the following two power systems:
System A: 6×50 MW Units-FOR 5%, Peak load of 250 MW
System B: 6×100 MW Units-FOR 6%, Peak load of 500 MW

The annual daily peak load variation curve is given by a straight line from 100% to 40% points.

(a) Calculate the LOLE, LOEE and EIR of the System B without any assistance from System A. **(4 marks)**

(b) Now assume that the two power systems are connected by two tie lines, with each line rating of 50MW. A contractual arrangement exists between the two power systems, which is defined by "for System-B firm purchase of 50 MW is backed by the complete System-A". Under this contractual arrangement calculate the LOLE of System B, when the tie lines have availability of 96%. **(6 marks)**

Question B3 **(Total marks 10)**

Consider a single line representation of a radial three-phase power system as shown in Figure 1. Assume that any fault, single phase or otherwise, will trip all three phases. It is normally found in practice that lines and cables have a failure rate, which is approximately proportional to their lengths. For this example, let us assume that the main feeder (sections 1, 2, 3, and 4) have a failure rate of 0.125 f/km/yr and the lateral distributors (a, b, c, and d) have a failure rate of 0.25 f/km/yr. Assume that the repair time *r* for each of the main feeder is 3 hours and each of the lateral distributor is 2.5 hours.

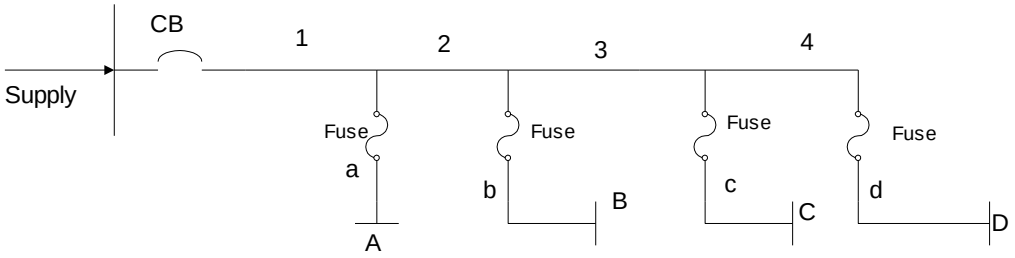


Figure 1 for Question B3

Reliability parameters for the system of Figure 1

Component	Length (km)
Section- 1	4
Section- 2	2
Section- 3	6
Section- 4	3
Distributor -a	2
Distributor -b	5
Distributor -c	2
Distributor -d	1

Customers and load connected to System in Figure 1

Load point	Number of customers	Average load connected (kW)
A	1200	5000
B	800	4000
C	800	3000
D	600	2500

(a) Assume that there is only one main CB. Assume that all component failures are short circuits, and then each failure will cause the main breaker to operate. For the time being assume that the other fuses do not exist. This means that there are no points at which the system can be isolated then each failure must be repaired before the breaker can be reclosed. On this basis calculate distribution system indices SAIFI, SAIDI, CAIDI, ASUI and ENS for this system assuming the reliability of each load point is identical. **(5 marks)**

(b) Let us now install fuse at the tee-point in each lateral distributor. In this case a short circuit on a lateral distributor causes its appropriate fuse to blow; this causes disconnection of its load point until the failure is repaired but does not affect or cause the disconnection of any other load point. On this basis calculate distribution system indices SAIFI, SAIDI, CAIDI, ASUI and ENS for this system. **(5 marks)**

END OF EXAMINATION

FORMULAE NEEDED FOR ANSWERING THE QUESTIONS ARE IN NEXT PAGES

FORMULAE – PART A

Variable names and meaning

P_0	Present value
F_n	Value in n years
A	Annuity (constant)
A_n	Annuity in year n
i	Interest rate
B_{jt}	Profit of plan j in year t
C_{jt}	Operation cost of plan j in year t
K_{jt}	Investment of plan j in year t
j	Number of the plan in consideration
n_j	Horizon of study (service/lifetime) of plan j
x_g	Electricity selling price of the grid
x_{FT}	Feed-in-tariff price
x_{MP}	Mid-market price
π	Marginal cost of the generator
P	Power produced by the generator
R	Revenue to the generator
PC	Production cost to the generator
D	Total demand of the customer
J	Cost per hour for customers

Present value

$$P_0 = \frac{F_n}{(1+i)^n}$$

Present Value Factor $(P|F,i,n) = \frac{1}{(1+i)^n}$

Future value

$$F_n = (1+i)^n \,.P_0$$

Future Value Factor $(F|P,i,n) = (1+i)^n$

Constant Annuity

$$F_n = A\frac{(1+i)^n - 1}{i} = A(F|A,i,n)$$

$$A = F_n\frac{i}{(1+i)^n - 1} = F_n(A|F,i,n)$$

$$P_0 = \frac{F_n}{(1+i)^n} = A\frac{(1+i)^n - 1}{i \cdot (1+i)^n} = A\frac{1-(1+i)^{-n}}{i} = A(P|A,i,n)$$

$$A = P_0\frac{i \cdot (1+i)^n}{(1+i)^n - 1} = P_0\frac{i}{1-(1+i)^{-n}} = P_0(A|P,i,n)$$

Capital Recovery Factor

$$CRF_j = \frac{i \cdot (1+i)^{n_j}}{(1+i)^{n_j} - 1} = \frac{i}{1-(1+i)^{-n_j}} = (A|P,i,n_j)$$

Incremental Annuity

$$P_0 = \sum_{k=1}^{n} A_k(1+i)^{-k} = \sum_{k=1}^{n} A_k(P|F,i,k)$$

Net present value

$$NPV_j = \sum_{t=1}^{n_j}\left[B_{jt}(P|F,i,t)\right] - \sum_{t=1}^{n_j}\left[(C_{jt}+K_{jt})(P|F,i,t)\right]$$

Benefit/Cost Ratio

$$BCR_j = \frac{\sum_{t=1}^{n_j}\left[(B_{jt}-C_{jt})(P|F,i,t)\right]}{\sum_{t=1}^{n_j}\left[K_{jt}(P|F,i,t)\right]}$$

Present Value of Costs

$$PVC_j = \sum_{t=1}^{n_j}\left[\left(C_{jt}+K_{jt}\right)(P|F,i,t)\right]$$

Mid-market Price (Price set via mid-market rule)

$$x_{MP} = \frac{x_g + x_{FT}}{2}$$

Capacity Factor (CF)

$$CF = \frac{kWh\ produced\ in\ a\ year}{rated\ power\ (kW) \times 8760h/year}$$

Revenue to generator per hour

$$R(P) = \pi \cdot P$$

Production cost to generator per hour

$$PC(P) = \int_0^{P_B} \pi_B dP$$

Cost per hour for customers

$$J = \pi \cdot D$$

Formulae-PART B

General

Failure rate/Forced Outage Rate: λ

Repair Rate: μ

Mean Time Before Failure (MTBF) $\frac{1}{\lambda}$

Mean Time To Repair (MTTR): $r = \frac{1}{\mu}$

For Generation and Composite Power System Reliability Assessment

Availability: $A = \frac{\mu}{\mu+\lambda}$

Unavailability: $U = \frac{\lambda}{\mu+\lambda}$

For Distribution Reliability Assessment

Availability: $A = 1$

Unavailability: $B = \lambda \cdot r = \frac{\lambda}{\mu}$

Common Customer Reliability Indices

System Average Interruption Frequency Index

$$\text{SAIFI} = \frac{\text{total number of customer interruptions}}{\text{total number of customers served}} = \frac{\sum N_i \cdot F_i}{\sum N_i}$$

System Average Interruption Duration Index

$$\text{SAIDI} = \frac{\text{sum of all customer interruption duration}}{\text{total number of customers served}} = \frac{\sum N_i \cdot Q_i}{\sum N_i}$$

Customer Average Interruption Frequency Index

$$\text{CAIFI} = \frac{\text{total number of customer interruptions}}{\text{total number of customers affected}} = \frac{\sum N_i \cdot F_i}{\sum N_i}$$

Customer Average Interruption Duration Index

$$\text{CAIDI} = \frac{\text{sum of all customer interruption duration}}{\text{total number of customer interruptions}} = \frac{\sum N_i \cdot Q_i}{\sum N_i \cdot F_i}$$

Average Service Availability Index

$$\text{ASAI} = \frac{\text{customer hours of available service}}{\text{customer hours demanded}} = \frac{\sum N_i \cdot (1 - Q_i)}{\sum N_i}$$

Average Service Unavailability Index $\text{ASUI} = 1 - \text{ASAI}$

Energy Not Supplied

$$\text{ENS} = \sum P_i \cdot Q_i \cdot 8760 \frac{\text{hrs}}{\text{yr}}$$

Average Energy Not Supplied

$$\text{AENS} = \frac{\text{total energy not supplied}}{\text{numbers of customers served}} = \frac{\sum P_i \cdot Q_i \cdot 8760 \frac{\text{hrs}}{\text{yr}}}{\sum N_i}$$

Average Curtailment Index

$$\text{ACCI} = \frac{\text{total energy not supplied}}{\text{number of customers affected}} = \frac{\sum P_i \cdot Q_i \cdot 8760 \frac{\text{hrs}}{\text{yr}}}{\sum N_i}$$